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Wireless data service using content-addressing

Abstract

A wireless communication system delivers a wireless data service to a wireless communication device using content-addressing. The wireless communication system generates a content-address from a data set. The wireless communication system stores the content-address in association with the data set. The wireless communication system encrypts the content-address with a system key and transfers the encrypted content-address to the wireless communication device. The wireless communication device decrypts the encrypted content-address with a user key to obtain the content address. The wireless communication device re-encrypts the content-address with the user key. The wireless communication system receives the re-encrypted content-address from the wireless communication device and decrypts the re-encrypted content-address with the system key to obtain the content-address. The wireless communication system retrieves the data set from the memory circuitry with the decrypted content-address. The wireless communication system delivers the wireless data service to the wireless communication device based on the retrieved data set.

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Background/Summary

TECHNICAL BACKGROUND

(1) Wireless communication networks provide wireless data services to wireless communication devices like phones, computers, and other user devices. The wireless data services may include internet-access, data messaging, video conferencing, or some other data communication functionality. The wireless communication networks comprise wireless access nodes like Wireless Fidelity (WIFI) hotspots and Fifth Generation New Radio (5G NR) cell towers. The wireless communication networks also comprise network slices. The network slices have customized software that is tailored for their specific wireless data services. For example, an augmented reality device may use an Ultra-Reliable Low Latency Communication (URLLC) slice while a television may use an enhanced Mobile Broadband (eMBB) slice.

(2) Before a wireless network slice delivers a wireless data services to a wireless communication device, the wireless communication network and the wireless communication device, the network and the user device engage in an authentication process and a slice selection procedure. The authorization process typically requires the network and the user device to hash a user identity and a random number and then the network matches the two hashes to authenticate the user device. The slice selection process typically requires the user device to submit a slice type and the network to interact among multiple network elements to select and authorize a slice for the user device based on the slice type. The current authentication process and selection procedure for a wireless communication device to use wireless network slice is cumbersome.

(3) Distributed ledgers have multiple ledger nodes that perform ledger transactions in parallel. The ledger nodes validate a transaction when a consensus is reached among the nodes for the ledger transaction. The typical ledger transaction entails a smart contract that processes a data input to generate a data-output. For example, a ledger may process the data inputs of a current balance and an expenditure to generate the data output of a new balance. The distributed ledger nodes each store transaction data in data blocks that also include a hash of the previous data block. Thus, the data blocks are linked by the hashes and the transaction data is immutable.

(4) Content-addressable storage is a way to store data sets so they can be retrieved based on their content. In a content-addressable system, the content of a data set is processed through a cryptographic hash function to generate a unique content identifier-sometimes called a content

address. The data file is stored in association with the content identifier. The content identifier is then used to request and retrieve the data set. As long as the content of the data set remains the same, the content identifier also remains the same and can be used to access the file. When the content of the data set changes, a new content identifier is generated based on the changed data set. The data sets in a content-addressing system are typically distributed across multiple system nodes. The content-addresses are able to obtain a desired data set from any of the nodes. A popular form of content-addressing is Inter-Planetary File System (IPFS).

(5) Unfortunately, the wireless communication networks require cumbersome authentication processes and slice selection procedures to allow secure user access to their network slices and other data services. Moreover, the wireless communication networks fail to use distributed ledgers and content-addressing in an efficient and effective manner to allow secure user access to their network slices other data services.

Technical Overview

(6) In some examples, a wireless data service is delivered to a wireless communication device using content-addressing. A content-address is generated from a data set and stored in association with the data set. The content-address is encrypted with a system key and transferred to the wireless communication device. The wireless communication device decrypts the encrypted content-address with a user key to obtain the content address. The wireless communication device re-encrypts the content-address with the user key. The re-encrypted content-address is received from the wireless communication device and decrypted with the system key to obtain the content-address. The data set is retrieved with the decrypted content-address. The wireless data service is delivered to the wireless communication device based on the retrieved data set.

(7) In some examples, a wireless data service is delivered to a wireless communication device using an Inter-Planetary File System Identifier (IPFS ID). The IPFS ID is a content address that is generated from content that comprises wireless network slice information. The IPFS ID is stored in association with the wireless network slice information. The IPFS ID is encrypted with a system key and is transferred to the wireless communication device. The wireless communication device decrypts the encrypted IPFS ID with a user key to obtain the IPFS ID. The wireless communication device encrypts the IPFS ID with the user key. The re-encrypted IPFS ID is received from the wireless communication device and decrypted with the system key to obtain the IPFS ID. The wireless network slice information is retrieved with the decrypted IPFS ID. The wireless data service is delivered to the wireless communication device based on the retrieved wireless network slice information.

(8) In some examples, a wireless data service is delivered to a wireless communication device using content-addressing. Processing circuitry generates a content-address from a data set. Memory circuitry stores the content-address in association with the data set. The processing circuitry encrypts the content-address with a system key and transfers the encrypted content-address to the wireless communication device. The wireless communication device decrypts the encrypted content-address with a user key to obtain the content address. The wireless communication device re-encrypts the content-address with the user key. The processing circuitry receives the re-encrypted content-address from the wireless communication device and decrypts the re-encrypted content-address with the system key to obtain the content-address. The processing circuitry retrieves the data set from the memory circuitry with the decrypted content-address.

Communication circuitry delivers the wireless data service to the wireless communication device based on the retrieved data set.

Description

DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an exemplary wireless communication system to deliver a wireless data service to a wireless communication device using content-addressing.
- (2) FIG. 2 illustrates an exemplary operation of the wireless communication system to deliver the wireless data service to the wireless communication device using content-addressing.
- (3) FIG. 3 illustrates an exemplary operation of the wireless communication system to deliver the wireless data service to the wireless communication device using content-addressing.
- (4) FIG. 4 illustrates exemplary processing circuitry to deliver a wireless data service to a wireless communication device using content-addressing.
- (5) FIG. 5 illustrates an exemplary wireless communication network to deliver a wireless data service to wireless User Equipment (UEs) using Inter-Planetary File System Identifiers (IPFS IDs).
- (6) FIG. 6 illustrates an exemplary UE in the wireless communication network that delivers the wireless data service to the wireless UE using an IPFS ID.
- (7) FIG. 7 illustrates an exemplary WIFI access node in the wireless communication network to deliver the wireless data service to the wireless UEs using the IPFS IDs.
- (8) FIG. 8 illustrates an exemplary Fifth Generation New Radio (5G NR) access node in the wireless communication network to deliver the wireless data service to the wireless UEs using the IPFS IDs.
- (9) FIG. 9 illustrates an exemplary data center in the wireless communication network to deliver the wireless data service to the wireless UEs using the IPFS IDs.
- (10) FIG. 10 illustrates an exemplary operation of the wireless communication network to deliver the wireless data service to a wireless UE using an IPFS ID.

DETAILED DESCRIPTION

- (11) FIG. 1 illustrates exemplary wireless communication system **100** to deliver a wireless data service to wireless communication device **101** using content-addressing. Wireless communication system **100** comprises wireless communication device **101**, communication circuitry **111**, processing circuitry **112**, and memory circuitry **113**. Wireless communication system **100** serves wireless data services to wireless communication device **101** like internet-access, data messaging, media conferencing, or some other data communications product. The amount of wireless communication devices that are shown in FIG. 1 has been restricted for clarity.
- (12) Wireless communication system **100** delivers the wireless data service to wireless communication device **101** using content-addressing. Processing circuitry **112** generates a content-address from a data set. For example, processing circuitry **112** may generate an Inter-Planetary File System Identifier (IPFS ID) from wireless network slice information. Processing circuitry **112** stores the content-address in association with the data set in memory circuitry **113**. Processing circuitry **112** encrypts the content-address with a system key. Processing circuitry **112** transfers the encrypted content-address to wireless communication device **101** over communication circuitry **111**. Wireless communication device **101** decrypts the encrypted content-address with a user key to obtain the content address.
- (13) When the wireless data service is needed, wireless communication device **101** re-encrypts the content-address with the user key. Processing circuitry **112** receives a re-encrypted content-address from wireless communication device **101** over communication circuitry **111**. Processing circuitry **112** decrypts the re-encrypted content-address with the system key to obtain the content-address. Processing circuitry **112** retrieves the data set from memory circuitry **113** using the decrypted content-address. Communication circuitry **111** delivers the wireless data service to wireless communication device **101** based on the retrieved data set. For example, communication circuitry **111** may deliver Ultra-Reliable Low-Latency Communication (URLLC) service to wireless communication device **101** based on a data set that comprises wireless network slice information for the URLLC slice.

(14) The data set may comprise wireless network slice information for the wireless communication service delivered to the wireless communication device. The data set may comprise operating system container properties for the wireless data service delivered to the wireless communication device. Processing circuitry **112** may comprise a smart contract in a distributed ledger that generates the content-address from the data set. Memory circuitry **113** may comprise a distributed ledger that stores the content-address in association with the data set.

(15) Wireless communication device **101** comprises a phone, computer, vehicle, sensor, or some other user communication apparatus. Communication circuitry **111** comprises wireless access nodes, network controllers, data routers, and/or some other wireless communication apparatus. Wireless communication device **101** and communication circuitry **111** comprise one or more radios that wirelessly communicate using wireless protocols like Wireless Fidelity, (WIFI), Fifth Generation New Radio (5G NR), Long Term Evolution (LTE), Low-Power Wide Area Network (LP-WAN), Near-Field Communications (NFC), Code Division Multiple Access (CDMA), Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), and satellite data communications.

(16) Wireless communication device **101** and circuitry **111-113** comprise microprocessors, software, memories, transceivers, bus circuitry, and/or some other data processing components. The microprocessors comprise Digital Signal Processors (DSP), Central Processing Units (CPU), Graphical Processing Units (GPU), Application-Specific Integrated Circuits (ASIC), and/or some other data processing hardware. The memories comprise Random Access Memory (RAM), flash circuitry, disk drives, and/or some other type of data storage. The memories store software like operating systems, utilities, protocols, applications, and functions. The microprocessors retrieve the software from the memories and execute the software to drive the operation of wireless communication system **100** as described herein.

(17) FIG. 2 illustrates an exemplary operation of wireless communication system **100** to deliver the wireless data service to wireless communication device **101** using content-addressing. The operation may differ in other examples. Wireless communication system **100** generates a content-address from a data set and stores the content-address in association with the data set (**201**). Wireless communication system **100** encrypts the content-address with a system key and transfers the encrypted content-address to wireless communication device **101** (**202**). Wireless communication device **101** decrypts the encrypted content-address with a user key to obtain the content address (**202**). Subsequently, wireless communication device **101** re-encrypts the content-address with the user key (**203**). Wireless communication system **100** receives a re-encrypted content-address from wireless communication device **101** (**203**). Wireless communication system **100** decrypts the re-encrypted content-address with the system key to obtain the content-address (**204**). Wireless communication system **100** retrieves the data set from memory circuitry **113** with the decrypted content-address (**204**). Wireless communication system **100** delivers the wireless data service to wireless communication device **101** based on the retrieved data set (**205**).

(18) FIG. 3 illustrates an exemplary operation of wireless communication system **100** to deliver the wireless data service to wireless communication device **101** using content-addressing. The operation may differ in other examples. Processing circuitry **112** generates a content-address from a data set. Processing circuitry **112** stores the content-address in association with the data set in memory circuitry **113**. Processing circuitry **112** encrypts the content-address with a system key. Processing circuitry **112** transfers the encrypted content-address to wireless communication device **101** over communication circuitry **111**. Wireless communication device **101** decrypts the encrypted content-address with a user key to obtain the content address. Wireless communication device **101** identifies (IDs) the need for the wireless data service, and in response, wireless communication device **101** re-encrypts the content-address with the user key. Processing circuitry **112** receives the re-encrypted content-address from wireless communication device **101** over communication circuitry **111**. Processing circuitry **112** decrypts the re-encrypted content-address with the system

key to obtain the content-address. Processing circuitry **112** retrieves the data set from memory circuitry **113** using the decrypted content-address. Communication circuitry **111** delivers the wireless data service to wireless communication device **101** based on the retrieved data set by exchanging user data between wireless communication device **101** and an external system (not shown).

(19) Advantageously, wireless communication system **100** avoids cumbersome authentication processes and slice selection procedures to allow secure user access to network slices and other data services. Moreover, wireless communication system **100** can use distributed ledgers and content-addressing in an efficient and effective manner to allow secure user access to the network slices and other data services.

(20) FIG. 4 illustrates exemplary processing circuitry **400** to deliver a wireless data service to a wireless communication device using content-addressing. Processing circuitry **400** comprises an example of circuitry **111-113**, although circuitry **111-113** may differ. Processing circuitry **400** comprises machine-readable storage media **401-403** and microprocessors **407-409** that are communicatively coupled. Machine-readable storage media **401-403** store processing instructions **404-406** in a non-transitory manner. Microprocessors **407-409** comprise DSPs, CPUs, GPUs, ASICs, and/or some other data processing hardware. Machine-readable storage media **401-403** comprises RAM, flash circuitry, disk drives, and/or some other type of data storage apparatus. Microprocessors **407-409** retrieve processing instructions **404-406** from non-transitory machine-readable storage media **401-403**. Microprocessors **407-409** execute processing instructions **404-406** to deliver the wireless data service to the wireless communication device using content-addressing as described above for wireless communication system **100** and as described below for wireless communication network **500**. The amount of storage media, microprocessors, processing instructions that are shown in FIG. 4 is exemplary and may vary in other examples.

(21) FIG. 5 illustrates exemplary wireless communication network **500** to deliver a wireless data service to wireless User Equipment (UEs) **501-503** using Inter-Planetary File System Identifiers (IPFS IDs). Wireless communication network **500** comprises an example of wireless communication system **100** and processing circuitry **400**, although system **100** and circuitry **400** may differ. Wireless communication network **500** comprises UEs **501-503**, Fifth Generation New Radio Access Node (5G NR AN) **504**, WIFI AN **505**, and Data center **530**. Data center **530** comprises Non-Third Generation Partnership Project Interworking Function (IWF) **506**, Access and Mobility Management Function (AMF) **507**, Session Management Function (SMF) **508**, slice **509**, and distributed ledger node **520**. Slice **509** comprises User Plane Function (UPF) **510**. Distributed ledger node **520** communicates with other distributed ledger nodes **521** and comprises smart contract **522** and ledger block **523**. Ledger block **523** comprises IPFS ID **524**, slice information (INFO) **525** and previous block hash **526**. Smart contract **522** includes a network key and UE **503** has a UE key (not shown). The network key and the UE key form a private/public key pair that is suitable for cryptography. The amount of UEs, ANs, data centers, slices, and ledger blocks has been restricted for clarity.

(22) In operation, distributed ledger node **520** executes smart contract **522** which has a network key. Smart contract **522** receives slice information **525** for UE **503** from network operations (not shown). Slice information **525** indicates Network Slice Selection Assistance Information (NSSAI), Slice/Service Types (SST), Slice Differentiator (SDs), quality-of-service levels, operating system container parameters and/or some other slice information. The quality-of-service levels may specify data throughput, data latency, error rate, and the like. The operating system container parameters may specify data Central Processing Unit (CPU) occupancy, memory resources, Input/Output (I/O) capability and/or some other container features.

(23) Smart contract **522** generates IPFS ID **524** from the content of slice information **525**. Smart contract **522** obtains consensus on IPFS ID **524** and slice information **525** for UE **503** from distributed ledger nodes **521**. Smart contract **522** stores IPFS ID **524** and slice information **525** in

ledger block **523**. Smart contract **522** stores the data from any previous blocks in block hash **526**—if any previous ledger blocks exist.

(24) Smart contract **522** encrypts IPFS ID **524** with the network key. In some examples, smart contract **522** transfers encrypted IPFS ID **524** to UE **503** over AMF **507** and 5G NR AN **504**. In other examples, smart contract **522** transfers encrypted IPFS ID **524** to UE **503** over AMF **507**, IWF **506** and WIFI AN **505**. AMF **507** may transfer encrypted IPFS ID **524** to UE **503** in a Non-Access Stratum (NAS) file over an N1 link that traverses 5G NR AN **504** or the combination of IWF **506** and WIFI AN **505**. UE **503** decrypts encrypted IPFS ID **524** with its UE key that is paired with the network key.

(25) When UE **503** determines that the wireless data service is required, UE **503** encrypts IPFS **503** with the UE key and transfers encrypted IPFS ID **524** to smart contract **522** over 5G NR AN **504** and AMF **507** or over WIFI AN **505**, IWF **506**, and AMF **507**. UE **503** may transfer encrypted IPFS ID **524** to AMF **507** in a NAS file over the N1 link. Smart contract **522** decrypts encrypted IPFS ID **524** with the network key that is paired with the UE key. Smart contract **522** obtains slice information **525** based on IPFS ID **524** which shares ledger block **523** with slice information **525**. Smart contract **522** transfers slice information **525** to AMF **507**.

(26) AMF **507** selects network slice **509** based on slice information **525** and indicates network slice **509** and transfers slice information **525** to SMF **508**. SMF **508** selects UPF **510** based on network slice **509** and slice information **525**. SMF **508** directs UPF **510** to serve UE **503** based on slice information **525**. In particular, SMF **508** directs UPF **510** to signal the operating system in data center **530** that executes UPF **510** to use operating system container properties that are indicated by slice information **525**. SMF **508** also directs UPF **510** to use quality-of-service levels indicated by slice information **525**. AMF **507** directs either 5G NR AN **504** or IWF **506** to serve UE **503** based on slice information **525**. In particular, AMF **507** directs 5G NR AN **504** or IWF **506** use the quality-of-service levels indicated by slice information **525**. AMF **507** directs UE **503** to use the quality-of-service levels indicated by slice information **525**. UE **503** exchanges user data with an external system (not shown) over slice **509** (UPF **510**) using 5G NR AN **505** or WIFI AN **505** and IWF **506**.

(27) In a similar manner slice information **525**, smart contract **522** may transfer software, non-fungible tokens, media, enterprise virtual states, or other items to UE **503**. For example, smart contract **522** may store a non-fungible token for UE **503** in association with IPFS ID **524** instead of storing slice information **525**. Smart contract **522** may then transfer the non-fungible token to UE **503** in response to receiving encrypted IPFS ID **524** as described above for slice information **525**. In some examples a Network Slice Selection Function (NSSF) or a Network Exposure Function (NEF) hosts smart contract **522**, and IPFS ID **524** indicates that NSSF or NEF.

(28) In some examples, a series of different IPFS IDs may be generated from a series of changing slice information, and the series of IPFS IDs and slice information could be stored in a series of ledger blocks that form an association between all of the IPFS IDs and slice information. For example, the smart contract could add slice usage data to modify slice information **525** and generate a new IPFS ID based on the modified slice information. The new IPFS ID could be issued to the user used to access the wireless data service over the wireless network slice. Ledger block hash **526** would record this usage history for the user of the wireless network slice and could be used for accounting.

(29) FIG. 6 illustrates exemplary UE **503** in wireless communication network **500** that delivers the wireless data service using IPFS ID **524**. UE **503** comprises an example of wireless communication device **101** and UEs **501-502**, although device **101** and UEs **502-503** may differ. UE **503** comprises WIFI radio circuitry **601**, 5G NR radio circuitry **602**, processing circuitry **603**, and components **604**. Components **604** comprise sensors, cameras, medical devices, and/or some other user apparatus. Radios **601-602** comprise antennas, amplifiers, filters, modulation, analog-to-digital interfaces, DSPs, memories, and transceivers (XCVRs) that are coupled over bus circuitry. Processing

circuitry **603** comprises one or more CPUs, one or more memories, and one or more transceivers that are coupled over bus circuitry. The one or more memories in processing circuitry **603** store a UE key and software like an Operating System (OS), 5G NR application (5G NR), 3GPP application (3GPP), Internet Protocol application (IP), and WIFI application (WIFI). The antennas in WIFI radio circuitry **601** exchange WIFI signals with WIFI AN **505**. The antennas in 5G NR radio circuitry **602** exchange 5G NR signals with 5G NR AN **504**. Transceivers in radios **601-602** are coupled to transceivers in processing circuitry **603**. In processing circuitry **603**, the one or more CPUs retrieve the software from the one or more memories and execute the software to direct the operation of UE **503** as described herein.

(30) In particular, radios **601-602** wirelessly receives encrypted IPFS ID **524** that was transferred by smart contract **522** for delivery to UE **503**. Processing circuitry **603** decrypts encrypted IPFS ID **524** with its UE key. When the wireless data service is required, processing circuitry **603** encrypts IPFS ID **524** with the UE key and radios **601-602** wirelessly transfer encrypted IPFS ID **524** for delivery to smart contract **522**. UE **503** may receive and transfer IPFS ID **524** in a NAS file over an N1 link. UE **503** uses the QoS levels indicated by slice information **525**. UEs **501-502** could be configured and operate like UE **503** where UE **501** uses 5G NR AN **504** and UE **502** uses WIFI AN **505**. UEs **501-502** could have different UE keys and use different slices than UE **503**.

(31) FIG. 7 illustrates exemplary WIFI access node **505** in wireless communication network **500** to deliver the wireless data service to wireless UEs **502-503** using IPFS IDs. WIFI AN **505** comprises an example of communication circuitry **111**, although circuitry **111** may differ. WIFI AN **505** comprises WIFI radio **701** and processing circuitry **702**. Radio **701** comprises antennas, amplifiers, filters, modulation, analog-to-digital interfaces, DSPs, memories, and transceivers that are coupled over bus circuitry. Processing circuitry **702** comprises one or more CPUs, one or more memories, and one or more transceivers that are coupled over bus circuitry. The one or more memories in processing circuitry **702** store software like an Operating System, WIFI application, and IP application. The antennas in WIFI radio **701** exchange WIFI signals with UEs **502-503**. Transceivers in radio **701** are coupled to transceivers in processing circuitry **702**. Transceivers in processing circuitry **702** are coupled to transceivers in IWF **506**. In processing circuitry **702**, the one or more CPUs retrieve the software from the one or more memories and execute the software to direct the operation of WIFI AN **505** as described herein.

(32) FIG. 8 illustrates exemplary Fifth Generation New Radio (5G NR) Access Node (AN) in wireless communication network **500** to deliver the wireless data service to wireless UEs **501** and **503** using the IPFS IDs. 5G NR AN **504** comprises an example of communication circuitry **111**, although circuitry **111** may differ. 5G NR AN **504** comprises 5G NR Radio Unit (RU) **801**, Distributed Unit (DU) **802**, and Centralized Unit (CU) **803**. 5G NR RU **801** comprises antennas, amplifiers, filters, modulation, analog-to-digital interfaces, DSP, memory, radio applications, and transceivers that are coupled over bus circuitry. DU **802** comprises memory, CPU, user interfaces and components, and transceivers that are coupled over bus circuitry. The memory in DU **802** stores operating system and 5G NR network applications for Physical Layer (PHY), Media Access Control (MAC), and Radio Link Control (RLC). CU **803** comprises memory, CPU, and transceivers that are coupled over bus circuitry. The memory in CU **803** stores an operating system and 5G NR network applications for Packet Data Convergence Protocol (PDCP), Service Data Adaptation Protocol (SDAP), and Radio Resource Control (RRC). The antennas in 5G NR RU **801** are wirelessly coupled to UE **501** and UE **503** over 5G NR links. Transceivers in 5G NR RU **801** are coupled to transceivers in DU **802**. Transceivers in DU **802** are coupled to transceivers in CU **803**. Transceivers in CU **803** are coupled to AMF **507** and UPF **512**. The DSP and CPU in RU **701**, DU **702**, and CU **703** execute the radio applications, operating systems, and network applications to exchange data and signaling with UE **501**, UE **503**, AMF **507**, and UPF **512** as described herein. In particular, 5G NR AN **504** serves UEs **501** and **503** based on quality of service levels indicated by slice information **525**.

(33) FIG. 9 illustrates exemplary data center 530 in wireless communication network 500 to deliver the wireless data service to wireless UEs 501-503 using the IPFS IDs. Data center 530 comprises an example of circuitry 111-113 and circuitry 400, although circuitry 111-113 and circuitry 400 may differ. Data center 530 comprises NF hardware 901, NF hardware drivers 902, NF operating systems 903, NF virtual layer 904, and NF Software (SW) 905. NF hardware 901 comprises Network Interface Cards (NICs), CPU, RAM, Flash/Disk Drives (DRIVE), and Data Switches (DSW). NF hardware drivers 902 comprise software that is resident in the NIC, CPU, RAM, DRIVE, and DSW. NF operating systems 903 comprise kernels, modules, applications, and containers. NF virtual layer 904 comprises vNIC, vCPU, vRAM, vDRIVE, and vSW. NF SW 905 comprises IWF SW 906, AMF SW 907, SMF SW 908, UPF SW 910, and ledger node SW 920. The NIC in NF hardware 901 are coupled to 5G NR AN 504, WIFI AN 505, and external systems. NF hardware 901 executes NF hardware drivers 902, NF operating systems 903, NF virtual layer 904, and NF SW 905 to form and operate IWF 506, AMF 507, SMF 508, UDM 510, and ledger node 520. The drives in NF hardware 901 comprise the memory for ledger node SW 920. NF operating system 903 uses container properties indicated by slice information 525, and either SMF SW 908 or UPF SW 909 signals these container properties to NF operating systems 903. Thus, data center 530 comprises one or more microprocessors and one or more non-transitory machine-readable storage media that store processing instructions that direct data center 530 to perform the methods described herein. Network data center 530 may be located at a single site or be distributed across multiple geographic locations.

(34) In operation, ledger node SW 920 receives slice information 525 for UE 503 from network operations (not shown). Ledger node SW 920 generates IPFS ID 524 from the content of slice information 525. Ledger node SW 920 obtains consensus on IPFS ID 524 and slice information 525 for UE 503 from distributed ledger nodes 521. Ledger node SW 920 stores IPFS ID 524 and slice information 525 in ledger block 523. Ledger node SW 920 stores the data from previous blocks in block hash 526 (if any previous ledger blocks exist).

(35) Ledger node SW 920 encrypts IPFS ID 524 with the network key. In some examples, ledger node SW 920 transfers encrypted IPFS ID 524 to UE 503 over AMF SW 907 and 5G NR AN 504. In other examples, ledger node SW 920 transfers encrypted IPFS ID 524 to UE 503 over AMF SW 907, IWF SW 906 and WIFI AN 505. AMF SW 907 may transfer encrypted IPFS ID 524 to UE 503 in a NAS file over an N1 link that traverses 5G NR AN 504 or the combination of IWF SW 906 and WIFI AN 505. UE 503 transfers encrypted IPFS ID 524 to ledger node SW 920 over 5G NR AN 504 and AMF SW 907 or over WIFI AN 505, IWF SW 906, and AMF SW 907. UE 503 may transfer encrypted IPFS ID 524 to AMF SW 907 in a NAS file over the N1 link. Ledger node SW 920 decrypts encrypted IPFS ID 524 with the network key that is paired with the UE key. Ledger node SW 920 obtains slice information 525 based on IPFS ID 524 which shares ledger block 523 with slice information 525. Ledger node SW 920 transfers slice information 525 to AMF SW 907.

(36) AMF SW 907 selects network slice 509 based on slice information 525 and indicates network slice 509 and slice information 525 to SMF SW 908. SMF SW 908 selects UPF SW 910 based on network slice 509 and slice information 525. SMF SW 908 directs UPF SW 910 to serve UE 503 based on slice information 525. In particular, SMF SW 908 directs UPF SW 910 to signal NF operating systems 903 that executes UPF SW 910 to use operating system container properties that are indicated by slice information 525. SMF SW 908 also directs UPF SW 910 to use quality-of-service levels indicated by slice information 525. AMF SW 907 directs either 5G NR AN 504 or IWF SW 906 to serve UE 503 based on slice information 525. In particular, AMF SW 907 directs 5G NR AN 504 or IWF SW 906 use the quality-of-service levels indicated by slice information 525. AMF SW 907 directs UE 503 to use the quality-of-service levels indicated by slice information 525. UE 503 exchanges user data with an external system (not shown) over slice 509 (UPF SW 910) using 5G NR AN 505 or WIFI AN 505 and IWF SW 906.

(37) In a similar manner slice information 525, ledger node SW 920 may transfer software, non-

fungible tokens, media, enterprise virtual states, or other items to UE 503. For example, ledger node SW 920 may store a non-fungible token for UE 503 in association with IPFS ID 524 instead of storing slice information 525. Ledger node SW 920 may then transfer the non-fungible token to UE 503 in response to receiving encrypted IPFS ID 524 as described above for slice information 525. In some examples Network Slice Selection Function (NSSF) SW or Network Exposure Function (NEF) SW hosts ledger node SW 920, and IPFS ID 524 indicates that NSSF or NEF.

(38) FIG. 10 illustrates an exemplary operation of wireless communication network 500 to deliver the wireless data service to wireless UE 503 using IPFS ID 524. The operation may differ in other examples. Distributed ledger node 520 generates IPFS ID 524 from slice information 525.

Distributed ledger node 520 obtains consensus (DLX) on IPFS ID 524 and slice information 525 for UE 503 from distributed ledger nodes 521 (not shown on FIG. 10). Distributed ledger node 520 stores IPFS ID 524 and slice information 525 in ledger block 523. Distributed ledger node 520 encrypts IPFS ID 524 with the network key. Distributed ledger node 520 transfers encrypted IPFS ID 524 to UE 503 over AMF 507 and 5G NR AN 504. UE 503 decrypts encrypted IPFS ID 524 with its UE key.

(39) When UE 503 determines that the wireless data service is required (ID SRV RQ), UE 503 encrypts IPFS 503 with the UE key and transfers encrypted IPFS ID 524 to distributed ledger node 520 over 5G NR AN 504 and AMF 507. Distributed ledger node 520 decrypts encrypted IPFS ID 524 with the network key. Distributed ledger node 520 obtains slice information 525 based on IPFS ID 524 which shares ledger block 523 with slice information 525. Distributed ledger node 520 obtains consensus from ledger nodes 521 on slice information 525 for UE 503. Distributed ledger node 520 transfers slice information 525 to AMF 507.

(40) AMF 507 selects network slice 509 based on slice information 525 and indicates network slice 509 and slice information 525 to SMF 508. SMF 508 selects UPF 510 based on network slice 509 and slice information 525. SMF 509 directs UPF 510 in selected slice 509 to serve UE 503 based on slice information 525. AMF 507 directs 5G NR AN 504 to serve UE 503 based on slice information 525 and the quality-of-service levels therein. AMF 507 directs UE 503 to use the quality-of-service levels indicated by slice information 525. 5G NR AN 504 and UPF 510 deliver the wireless data service to UE 503 based on slice information 525. UE 503 exchanges UE data with an external system (not shown) over 5G NR AN 505 and UPF 510 per slice information 525.

(41) The wireless communication system circuitry described above comprises computer hardware and software that form special-purpose data communication circuitry to deliver a wireless data service to a wireless communication device using content-addressing. The computer hardware comprises processing circuitry like CPUs, DSPs, GPUs, transceivers, bus circuitry, and memory. To form these computer hardware structures, semiconductors like silicon or germanium are positively and negatively doped to form transistors. The doping comprises ions like boron or phosphorus that are embedded within the semiconductor material. The transistors and other electronic structures like capacitors and resistors are arranged and metallically connected within the semiconductor to form devices like logic circuitry and storage registers. The logic circuitry and storage registers are arranged to form larger structures like control units, logic units, and Random-Access Memory (RAM). In turn, the control units, logic units, and RAM are metallically connected to form CPUs, DSPs, GPUs, transceivers, bus circuitry, and memory.

(42) In the computer hardware, the control units drive data between the RAM and the logic units, and the logic units operate on the data. The control units also drive interactions with external memory like flash drives, disk drives, and the like. The computer hardware executes machine-level software to control and move data by driving machine-level inputs like voltages and currents to the control units, logic units, and RAM. The machine-level software is typically compiled from higher-level software programs. The higher-level software programs comprise operating systems, utilities, user applications, and the like. Both the higher-level software programs and their compiled machine-level software are stored in memory and retrieved for compilation and execution. On

power-up, the computer hardware automatically executes physically-embedded machine-level software that drives the compilation and execution of the other computer software components which then assert control. Due to this automated execution, the presence of the higher-level software in memory physically changes the structure of the computer hardware machines into special-purpose data communication circuitry to deliver the wireless data service to the wireless communication device using content-addressing.

(43) The above description and associated figures teach the best mode of the invention. The following claims specify the scope of the invention. Note that some aspects of the best mode may not fall within the scope of the invention as specified by the claims. Those skilled in the art will appreciate that the features described above can be combined in various ways to form multiple variations of the invention. Thus, the invention is not limited to the specific embodiments described above, but only by the following claims and their equivalents.

Claims

1. A method to deliver a wireless data service to a wireless communication device using content-addressing, the method comprising: generating a content-address from a data set and storing the content-address in association with the data set; encrypting the content-address with a system key and transferring the encrypted content-address to the wireless communication device, wherein the wireless communication device decrypts the encrypted content-address with a user key to obtain the content address; receiving a re-encrypted content-address from the wireless communication device and decrypting the re-encrypted content-address with the system key to obtain the content-address, wherein the wireless communication device re-encrypted the content-address with the user key; retrieving the data set with the decrypted content-address; and delivering the wireless data service to the wireless communication device based on the retrieved data set.
2. The method of claim 1 wherein the data set comprises wireless network slice information for the wireless communication service delivered to the wireless communication device.
3. The method of claim 1 wherein the data set comprises operating system container properties for the wireless data service delivered to the wireless communication device.
4. The method of claim 1 wherein the data set comprises: wireless network slice information for the wireless communication service delivered to the wireless communication device; and operating system container properties for the wireless data service delivered to the wireless communication device.
5. The method of claim 1 wherein generating the content-address from the data set comprises generating an Inter-Planetary File System Identifier (IPFS ID) from the data set.
6. The method of claim 1 wherein generating the content-address from the data set comprises a smart contract in a distributed ledger generating the content-address from the data set.
7. The method of claim 1 wherein storing the content-address in association with the data set comprises a distributed ledger storing the content-address in association with the data set.
8. The method of claim 1 wherein the data set comprises wireless network slice information for the wireless data service delivered to the wireless communication device.
9. The method of claim 1 wherein the data set comprises operating system container properties for the wireless data service delivered to the wireless communication device.
10. The method of claim 1 wherein the data set comprises: wireless network slice information for the wireless data service delivered to the wireless communication device; and operating system container properties for the wireless data service delivered to the wireless communication device.
11. The method of claim 1 wherein the content-address comprises an Inter-Planetary File System Identifier (IPFS ID).
12. The method of claim 1 wherein the processing circuitry comprises a distributed ledger smart contract.

13. The method of claim 1 wherein the communication circuitry comprises a wireless network slice.

14. A method to deliver a wireless data service to a wireless communication device using an Inter-Planetary File System Identifier (IPFS ID), the method comprising: generating the IPFS ID from wireless network slice information and storing the IPFS ID in association with the wireless network slice information; encrypting the IPFS ID with a system key and transferring the encrypted IPFS ID to the wireless communication device, wherein the wireless communication device decrypts the encrypted IPFS ID with a user key to obtain the IPFS ID; receiving a re-encrypted IPFS ID from the wireless communication device and decrypting the re-encrypted IPFS ID with the system key to obtain the IPFS ID, wherein the wireless communication device encrypted the IPFS ID with the user key; retrieving the wireless network slice information with the decrypted IPFS ID; and delivering the wireless data service to the wireless communication device based on the retrieved wireless network slice information.

15. The method of claim 14 wherein generating the Inter-Planetary File System Identifier (IPFS ID) from the wireless network slice information comprises a smart contract in a distributed ledger generating the IPFS ID from the wireless network slice information.

16. The method of claim 14 wherein storing the IPFS ID in association with the wireless network slice information comprises a distributed ledger storing the IPFS ID in association with the wireless network slice data set in a distributed ledger.

17. The method of claim 14 wherein encrypting the IPFS ID with the system key and decrypting the re-encrypted IPFS ID with the system key comprises a smart contract in a distributed ledger encrypting the IPFS ID with the system key and decrypting the re-encrypted IPFS ID with the system key.

18. The method of claim 14 wherein the wireless network slice information comprises operating system container properties.

19. The method of claim 14 wherein the wireless network slice information comprises quality-of-service parameters for the wireless data service.

20. A wireless communication system to deliver a wireless data service to a wireless communication device using content-addressing, the wireless communication system comprising: processing circuitry to generate a content-address from a data set; memory circuitry to store the content-address in association with the data set; the processing circuitry to encrypt the content-address with a system key and transfer the encrypted content-address to the wireless communication device, wherein the wireless communication device is to decrypt the encrypted content-address with a user key to obtain the content address; the processing circuitry to receive a re-encrypted content-address from the wireless communication device and decrypt the re-encrypted content-address with the system key to obtain the content-address, wherein the wireless communication device is to re-encrypt the content-address with the user key; the processing circuitry to retrieve the data set from the memory circuitry with the decrypted content-address; and communication circuitry to deliver the wireless data service to the wireless communication device based on the retrieved data set.
